

Table 2: **Interaction quality.** Overall quality in each environment and five complementary dimensions of assistance, evaluated with the full CookSim rubric.

Assistant	Overall quality			Rubric dimensions: three-environment mean				
	CookSim	VHSim	ScreenSim	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness
Frontier API models								
GPT-6 Astra	64	34	76	84	90	77	79	85
Gemini 3.1 Pro	28	31	80	74	79	71	60	77
Gemini 3.8 Flash	53	46	78	86	86	76	80	89
Real-time streaming API models								
Gemini 3.8 Live (polled)	47	42	83	88	81	72	68	90
GPT-Realtime 2.1 (polled)	41	37	77	85	87	74	78	64
Gemini 3.8 Live (native)	40	—	—	—	—	—	—	—
GPT-Realtime 2.1 (native)	50	—	—	—	—	—	—	—
Open-weight non-streaming VLMs								
Qwen3.8-27B	37	29	82	78	85	73	74	76
Molmo2-8B	12	17	16	69	40	37	26	42
InternVideo3-8B	21	18	23	67	52	41	31	60
GLM-4.6V-Flash	47	43	43	90	60	55	50	83
Kimi-VL-A3B-Thinking	6	12	30	54	53	42	38	39
Muse Glimmer	27	26	55	70	83	67	69	54
Open-weight streaming VLMs								
MiniCPM-o 4.5	15	16	31	64	53	39	45	49
Qwen3-Omni-30B	25	30	50	75	73	61	45	77
Proact-VL	16	16	12	73	22	17	19	38
StreamingVLM	24	23	27	82	25	29	21	25

Scores are on a 0–100 scale (run 1). Overall: full 16-item CookSim rubric with mandatory-failure gating, not the mean of the five displayed dimensions. Dimensions: equal-weight means across all three environments; unavailable for CookSim-only native configurations. Bold: highest score per column.